

Durga sai prasad Pragada

Associate Business Analyst (Python & Gen AI)

India | +91 7780158135 | durgapragada204@gmail.com | [LinkedIn](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Python & Generative AI Developer with **2+ years of experience** building AI-powered applications, backend services, and LLM-based solutions. Experienced in developing **AI chatbots, Agentic AI workflows, RAG systems, LangChain-based applications, MCP integrations, and Multi-agentic systems**. Strong hands-on experience with Python, FastAPI, **LLM integration, prompt engineering, REST APIs, WebSockets, STT/TTS, and AI orchestration**. Experienced in designing scalable backend architectures, integrating AI capabilities with enterprise systems, and developing production-oriented AI solutions.

TECHNICAL SKILLS

Programming : Python
AI / ML : LLMs (OpenAI, Groq), Prompt Engineering, NLP, Speech-to-Text (STT), Text-to-Speech (TTS), LangChain, RAG, MCP, Agentic AI
Backend : FastAPI, REST API Development, JWT Authentication, Pandas
Databases : MySQL, Redis
Tools & Platforms: Git, GitHub, GitLab, Render, Linux, AWS

PROFESSIONAL EXPERIENCE

Infosys Limited, Bangalore, India Associate Business Analyst — Python & Gen AI | Sept 2026 – Present

- Developing **AI-powered chatbot applications** using Python and Generative AI technologies for enterprise use cases.
- Working with **Agentic AI architectures** to build intelligent systems capable of reasoning, tool usage, and multi-step task execution.
- Developing LLM-based applications using **LangChain** for prompt management, workflow orchestration, tool integration, and AI application development.
- Working with **Model Context Protocol (MCP)** to integrate AI agents with external tools, services, and enterprise capabilities through standardized interfaces.
- Exploring and implementing **Multi-Agentic Frameworks** for building collaborative AI workflows where multiple specialized agents work together to solve complex tasks.
- Implementing **RAG (Retrieval-Augmented Generation)** pipelines to provide context-aware responses using enterprise knowledge sources.
- Developing and optimizing prompts for improving chatbot response quality, accuracy, and consistency.
- Building and integrating **REST APIs and backend services** to connect AI applications with enterprise systems.
- Integrating LLMs and AI services with external tools and applications to automate business workflows.
- Working with Python asynchronous programming and backend components to develop scalable AI applications.
- Collaborating with cross-functional teams to understand business requirements and translate them into scalable AI-driven solutions.
- Working with cloud technologies including **AWS** for AI application development and deployment.

Deepija Telecom Private Limited, Hyderabad, India [Sept-2024 to August 2026]

Project 1: ConVox Platform Management System

- Developed core backend services including **Data Ingestion APIs, Operation Status APIs, and multi-endpoint workflow orchestration**. Designed and implemented **MySQL-based reporting modules** to monitor user activity, operational metrics, and system performance.
- Improved system performance by **~60%** using **Redis caching** for asynchronous tasks and high-frequency message routing.
- Enhanced platform stability through **structured testing, debugging, and API validation** using Postman.
- Collaborated closely with **product and engineering teams** to translate business requirements into **scalable backend solutions**.

Project 2: Real-Time AI Integration with ConVox

- Built a **Python-based WebSocket bridge** and **bidirectional server** to enable real-time communication with third-party AI services. Implemented an **event-driven architecture** to manage session initialization, live data streaming, and workflow completion events.
- Developed **low-latency processing modules** using Python **async programming and multithreading**.
- Integrated **webhook-based JSON reporting** to capture workflow results and generate detailed post-processing logs.
- Increased system throughput and reliability using **Redis caching, thread synchronization, retry mechanisms, and robust exception handling**.

Personal Project 1: Centralized AI Automation Platform -In progress [My-Product-link](#)

- Developed a multi-tenant AI platform integrating Voice Bots, Chat Bots, and AI-driven Call Quality Analysis. Enabled automated customer interactions, conversation transcription, analytics, and real-time insights through a unified dashboard. Designed a scalable architecture supporting multiple clients, improving customer experience, operational efficiency, and call monitoring capabilities.
- **Key Technologies:** Python, FastAPI, WebSockets, AI/LLM Integration, Speech-to-Text, Text-to-Speech, MySQL, Redis, Nginx, LangChain, Agentic AI

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Project 2: AI Web-Based Chatbot: [AI Voice Chat](#)

- Developed a web-based AI chatbot for hospitals to provide users with information about hospital services, departments, doctors, facilities, and other hospital-related queries.
- Implemented RAG (Retrieval-Augmented Generation) to retrieve relevant information from hospital knowledge sources and generate context-aware responses.
- Integrated Groq and OpenAI LLMs for natural-language understanding and conversational response generation.
- Developed chatbot workflows to assist users with hospital and doctor information and appointment booking-related queries.
- Implemented prompt engineering and contextual conversation handling to improve response accuracy and consistency.
- Built backend services using Python and FastAPI to handle chatbot requests and AI processing.
- Designed the chatbot architecture to support integration with hospital systems and additional services in the future.

INTERNSHIP EXPERIENCE [Deepija Telecom Pvt Ltd, Hyderabad, India | Sept 2024 – Jan 2025]

- Developed and maintained data reporting modules, processing large-scale application data using **MySQL**.
- Designed and optimized database queries for efficient data retrieval and performance improvement.
- Built and tested backend APIs using **Python** and **FastAPI** for data processing and workflow execution.
- Worked on system-level integrations, improving reliability and handling of real-time data workflows.
- Progressed into **AI-based system integration**, connecting backend services with **LLM-driven** applications for intelligent automation

CERTIFICATIONS

- IBM Prompt Engineering Basics (Coursera)
- IBM Generative AI: Introduction and Applications (Coursera)

Relevant Skills Acquired

Prompt Engineering, Large Language Models (LLMs), Generative AI, AI Applications, AI Content Generation, Prompt Design, AI Solution Development.

EDUCATION:

- **Masters of Computer Applications (MCA) [2021 – 2023] | CGPA: 8.0 / 10 | B V Raju College – Bhimavaram, Andhra Pradesh, INDIA.**

STRENGTHS

- Strong problem-solving skills
- Clear understanding of AI workflows
- Quick learner in new AI tools
- Good communication & teamwork